KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2020
Department of Computer Science and Engineering
CSE 4111
Machine Learning

TIME: 1.5 hours

FULL MARKS: 120

N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) Differentiate learning and generalization in the contexts of machine learning. Which one is more desirable and why? (08)

   b) Define classification, regression and logistic regression in short. Differentiate between the concepts: classification and regression; and regression and logistic regression. (10)

   c) Using K-Means clustering, cluster the following objects for $K = 2$.

Object	Attribute 1	Attribute 2
A	10	150
B	150	10
C	X	Y
D	150	150

   The attribute values of object C are $X =$ $Y =$ (Last three digits of your Roll Number). (12)

2. a) Suppose bias weight of an artificial neuron is $-X$, where X is the last three digits of your Roll Number. Sketch two input AND logic and OR logic by setting appropriate values of other weights of the neurons. (10)

   b) What are the significant properties of XOR in the context of machine learning? Is it possible to construct XOR logic with a neuron taking input from developed AND logic and OR logic? Do it if possible; otherwise explain the reasons for inability. (10)

   c) What is the local gradient in Back Propagation (BP) algorithm? "Local gradient may be zero even error remains" – Justify the statement with an appropriate activation function. (10)

3. a) Why dimensionality reduction is important in machine learning? How does principal component analysis reduce dimensionality? (10)

   b) What is Bayes theorem? Shortly describe Naive Bayes Classifier. (10)

   c) Consider the label data points in the following table. (10)

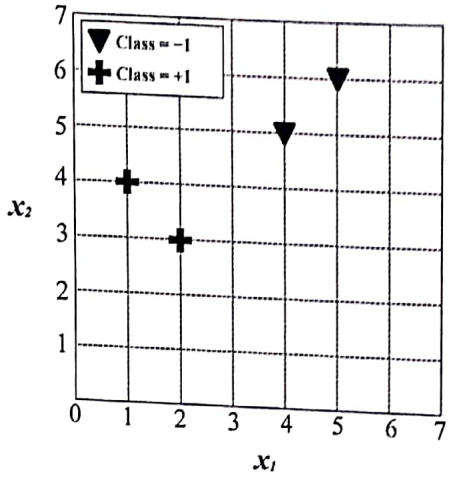
X_1	X_2	Label
05	15	+
10	30	−
30	40	+
40	100	−
80	130	+
120	250	−
Y_1	Y_2	?

   Identify the label of data point (Y_1, Y_2) using KNN for $K = 3$, where Y_1 = Last three digits of your Roll Number and $Y_2 = 2Y_1$.

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) Suppose you are training on a tiny dataset with 4 points shown in the following figure. This dataset consists of two examples with class label +1 (denoted with plus), and two examples with class label −1 (denoted with triangle). (10)



   i. Find the weight vector w and bias b. What is the equation corresponding to the decision boundary?
   ii. Circle the support vectors and draw the decision boundary.

   b) Consider the following training dataset, where A, B, and C are three binary attributes, and D is a binary class label. (20)

Instance	A	B	C	D
1	0	0	0	0
2	0	0	1	0
3	0	1	0	1
4	0	1	1	1
5	1	0	1	1
6	1	0	1	1
7	1	1	0	0
8	1	1	0	0

   i. What is the expected information (entropy) required to classify the training dataset?
   ii. Using ID3 algorithm, build a decision tree for this classification problem.
   iii. Use the built decision tree to classify two samples below.

Instance	A	B	C	D
9	1	1	1	?
10	1	0	0	?

5. a) "In PSO, each and every particle changes position in each iteration even the particle is the global best (G)" – do you agree with the statement? Justify your answer with appropriate assumption in velocity update equation. (09)

   b) Is it possible to find out the population size from the Roulette wheel given below? Explain. (07)



   c) Apply the Enhanced Edge Recommendation algorithm on the following two chromosomes to produce two children. Show all the steps. (14)

Parent 1	4	2	1	3	5	6
Parent 2	4	3	1	2	6	5

6. a) "Pheromone evaporation pushes ACO convergence" – Is it true? Justify your opinion. (10)

b) What two requirements should a problem satisfy in order to be suitable for solving it by GA? (05)

c) Assume 4 cities {A, B, C, D}, which are represented by a fully connected graph. The following tables represent the pheromone levels on each edge of the graph and the distances between each city (assume the pheromone levels and distances are symmetric). (15)

Table-1

Pheromone Levels				
	A	B	C	D
A				
B	0.25			
C	0.11	0.98		
D	0.34	0.54	0.67	

Table-2

Distance				
	A	B	C	D
A				
B	12			
C	10	6		
D	8	15	3	

Assume an ant started its journey at city A and has travelled to city C.

i. What are the probabilities that the ant will travel to cities A, B and D? [Let $\alpha = 1, \beta = 1$]

ii. Assume the ant completes its tour using the route ABCD. What will be the pheromone levels on each edge once they have been updated? [Let Q = 100 and evaporation parameter = 0.5]